Placard main board

Hierarchy: each box is one sheet; the sheets connect through global labels

Power and clock

J13 USB-C, F201, U40 3.3 V

X1 25.175 MHz, U41 clock gate

HOLD / RUN jumpers

File: power.kicad_sch

Control

U1 State, U2 Branch

U17 instruction register

File: control.kicad_sch

Program counter and address mux

U3, U4 program counter

U7–U10 address mux

File: pc_mux.kicad_sch

Memory and I/O decode

U31, U32 RAM, J8 cartridge

U11 memory control

U12 I/O decode, U25 map

File: memory.kicad_sch

Video timing

U13 H, U14 V counters

U15 Sync, U35 audio slots

U29 line read

File: video_timing.kicad_sch

ALU and registers

U5, U6 ALU

U19 AC, U20 X, U21 Y

File: alu_regs.kicad_sch

Data bus

U18 D register

U26, U27 source drivers

R601–R608, RN601

File: data_bus.kicad_sch

Expansion slots

J9–J12 card-edge slots

File: expansion.kicad_sch

Video and audio out

U22 pixel buffer, U23 line code

U16 color decoder, VGA

U36 audio DAC, J6

File: video_out.kicad_sch

Shared port

U37 latch, U38 port PLD

U39 rows, J1–J5

File: shared_port.kicad_sch

35 chips + U40 regulator (docs/CHIPS.md numbers; U24, U28, U30, U33, U34 are not used).
Decoupling: every chip's 100 nF Cn sits beside Un on its own sheet.
Power: +5V, GND, +3V3 (SD card), GNDA (audio, joined to GND on sheet 2).
The ROM cartridge is its own project: kicad/cartridge/.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

Placard computer

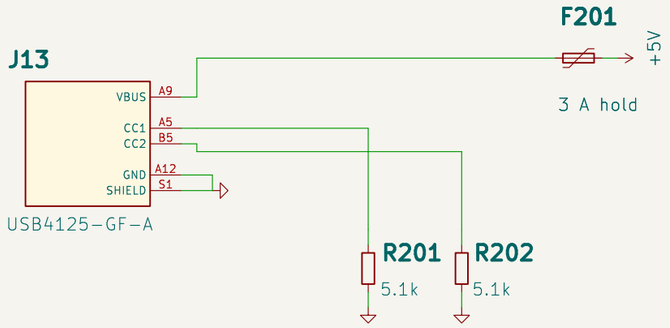
Sheet: /
File: placard.kicad_sch

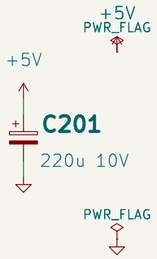
Title: Placard: Main board

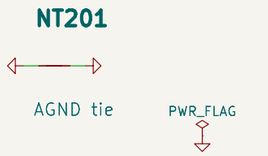
Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/11

Power and clock

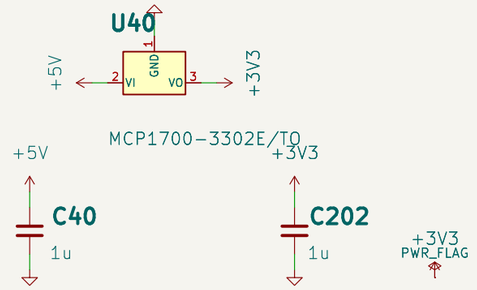
USB-C 5 V in, 3.3 V for the SD card, 25.175 MHz oscillator, clock gate CLK = OSC OR HOLD





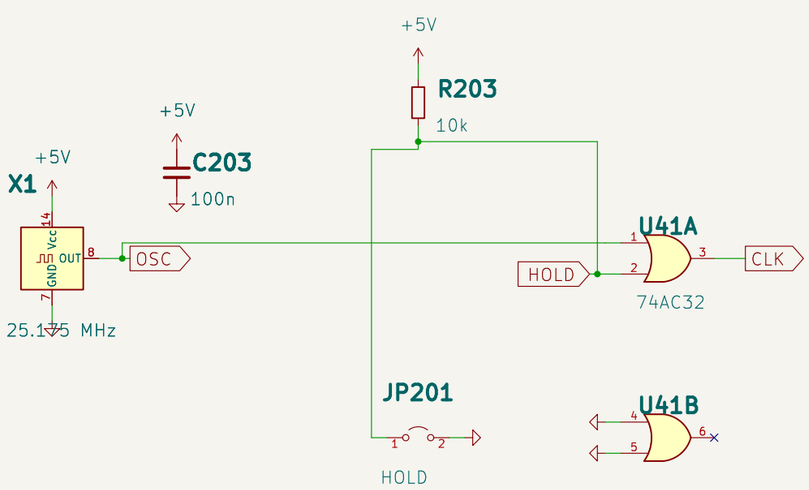


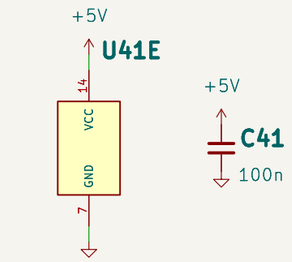
AGND (audio) meets GND here, at the power input

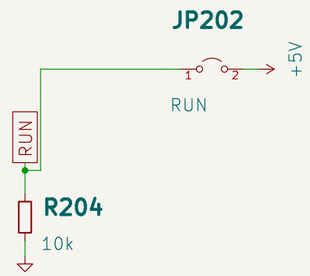


5.1k on CC1 and CC2: a USB-C source gives 5 V.
18 PLDs draw up to about 2.3 A (130 mA each at -7, worst case): use a 3 A supply; the polyfuse trips on a short.

U40: 3.3 V for the SD card only (sheet 10).
C40 input, C202 output (1 uF ceramic).







HOLD (10k up) and RUN (10k down) come from the cartridge. Without one, fit JP201 and JP202 for bring-up (BRINGUP.md); take them out with a cartridge.
Unused OR gates: inputs to GND.

Chip numbers U1-U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

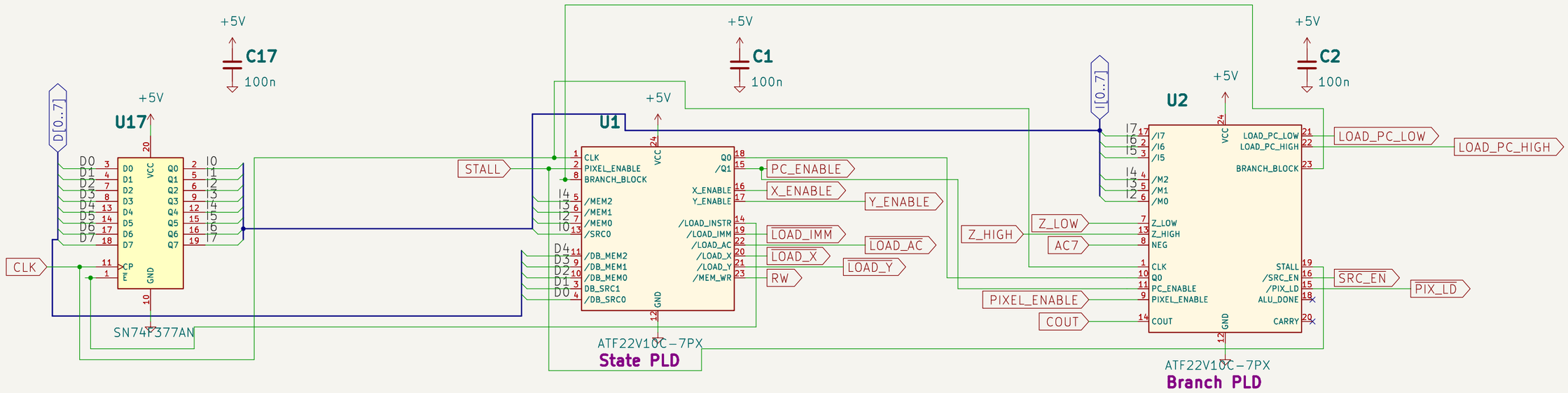
Sheet: /Power and clock/
File: power.kicad_sch

Title: Placard: Power and clock

Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/11

Control

State and Branch PLDs, instruction register



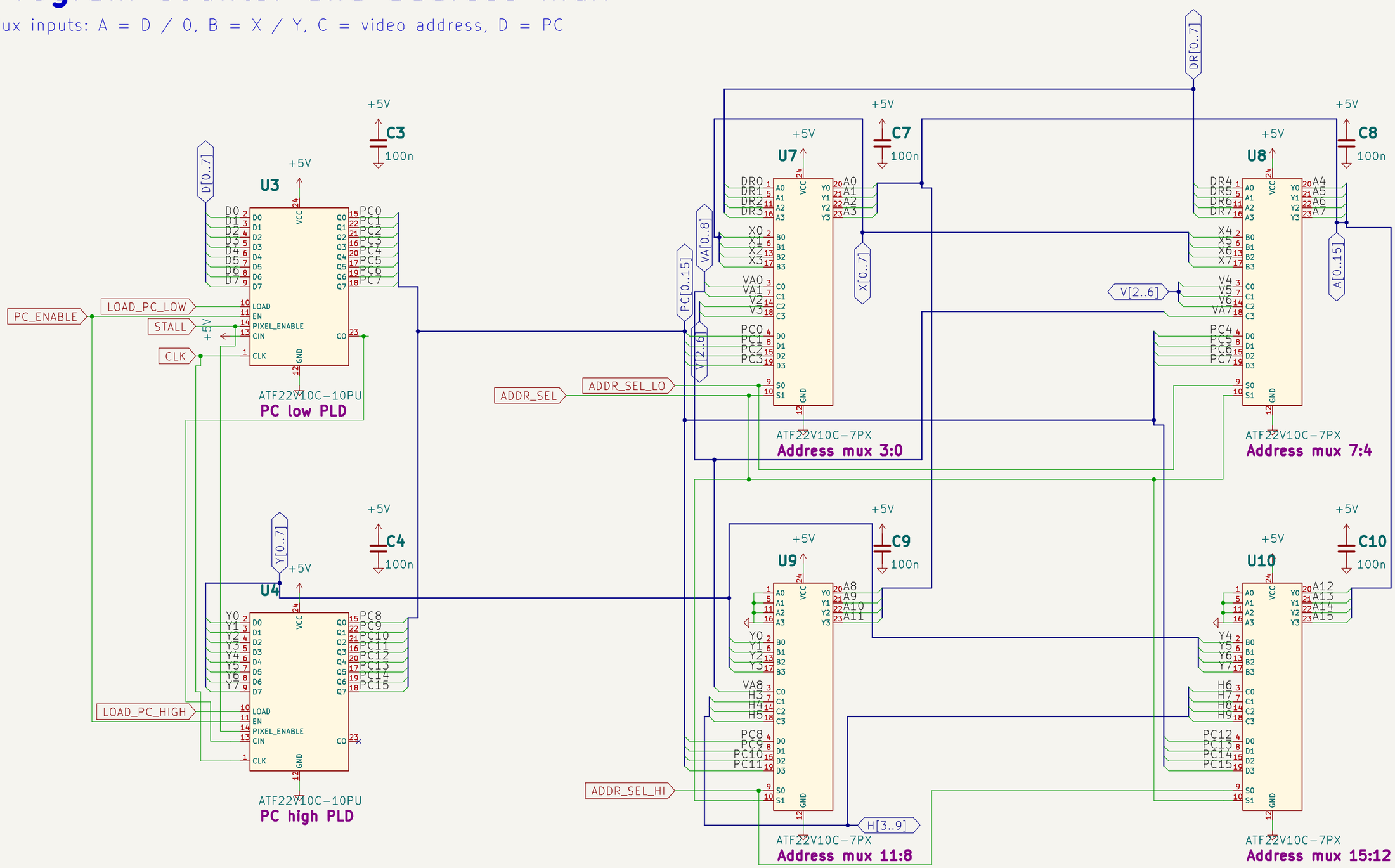
U17 loads the opcode in FETCH (/LOAD_INSTR). The byte is stored XOR FDh (CPU.md).
U1 State: FETCH / LOAD / EXEC, register loads (active low, F377 /E), RW (1 = read).
U2 Branch: NEG is AC7. ALU_DONE (18) and CARRY (20) are internal registers: no connection.
Bring-up straps S1-S3 go in U1's socket (BRINGUP.md), not on the board.

Chip numbers U1-U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Control/		
File: control.kicad_sch		
Title: Placard: Control		
Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/11

Program counter and address mux

Mux inputs: A = D / 0, B = X / Y, C = video address, D = PC



Video address: VA0, VA1, VA7, VA8 from the audio slot PLD (line repeat), A6:A2 = V6:V2, A15:A9 = H9:H3.
U3 CIN tied high (always counts); U3 CO carries into U4. STALL holds the PC during video reads and the ALU wait.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

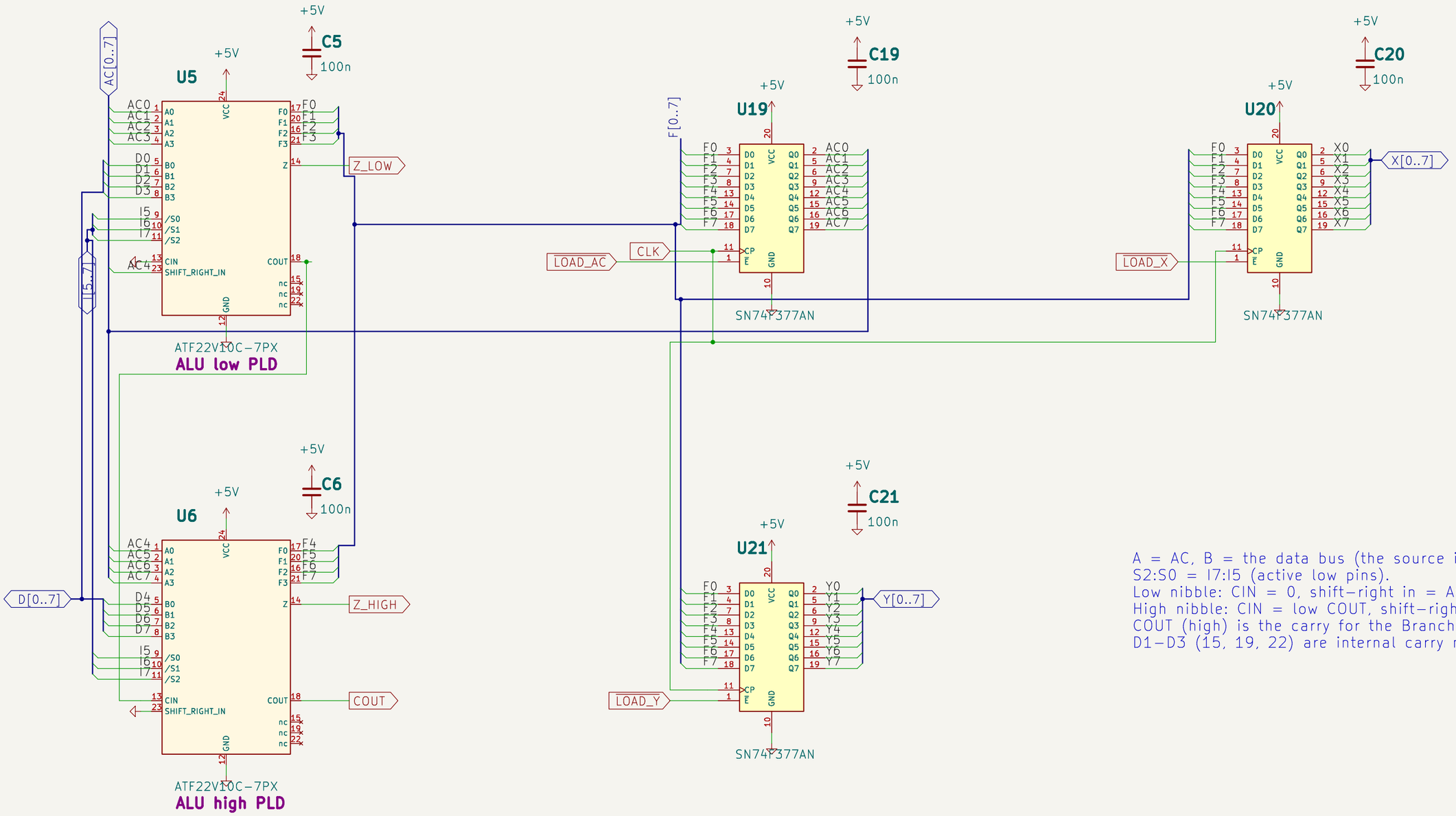
Sheet: /Program counter and address mux/
File: pc_mux.kicad_sch

Title: Placard: Program counter and address mux

Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/11

ALU and registers

Two 4-bit ALU PLDs; F7:F0 loads AC, X or Y



A = AC, B = the data bus (the source in EXEC).
S2:S0 = I7:I5 (active low pins).
Low nibble: CIN = 0, shift-right in = AC4.
High nibble: CIN = low COUT, shift-right in = 0.
COUT (high) is the carry for the Branch PLD.
D1-D3 (15, 19, 22) are internal carry nodes.

Chip numbers U1-U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

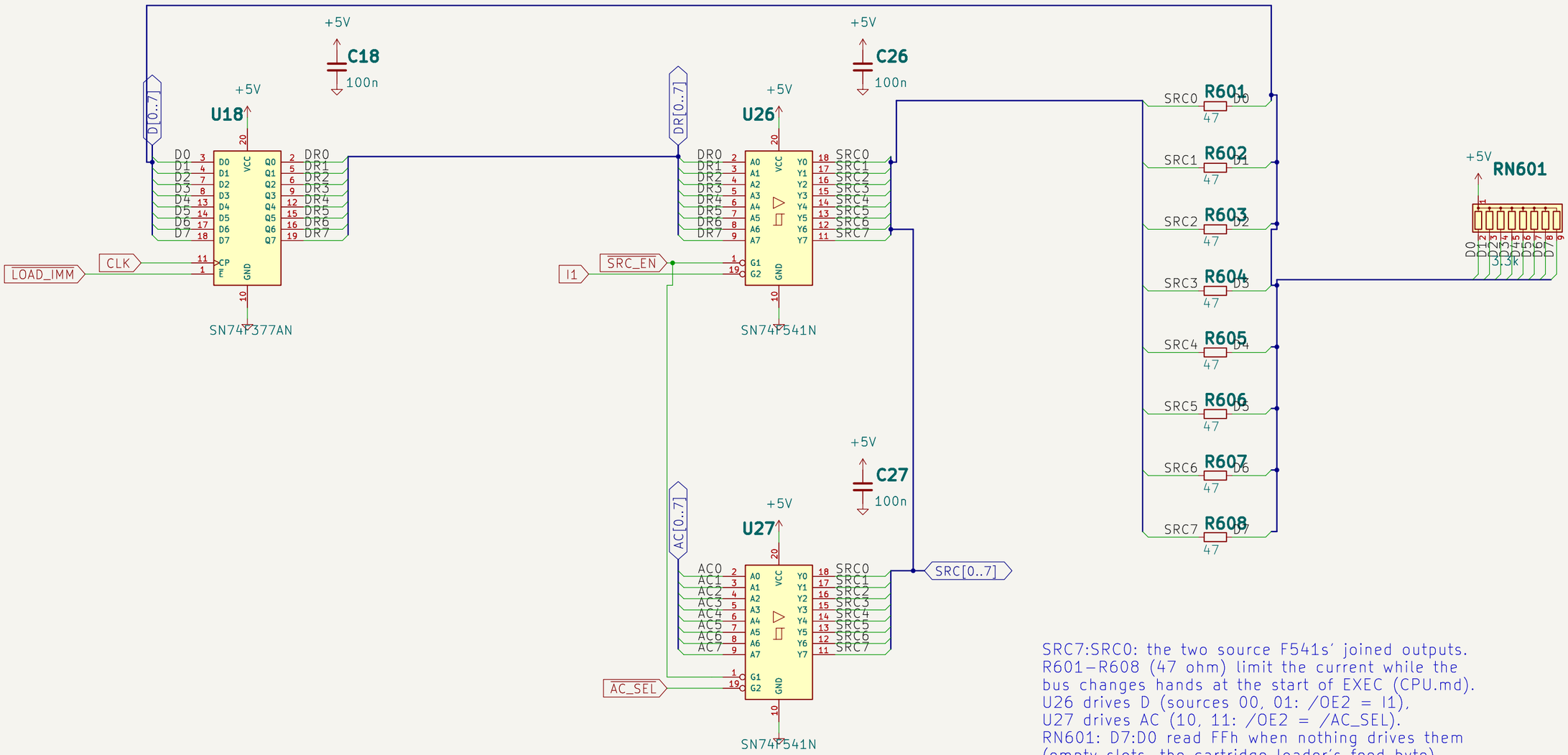
Sheet: /ALU and registers/
File: alu_regs.kicad_sch

Title: Placard: ALU and registers

Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/11

Data bus

D register, source drivers (D or AC) onto the one data bus through series resistors, read pull-ups



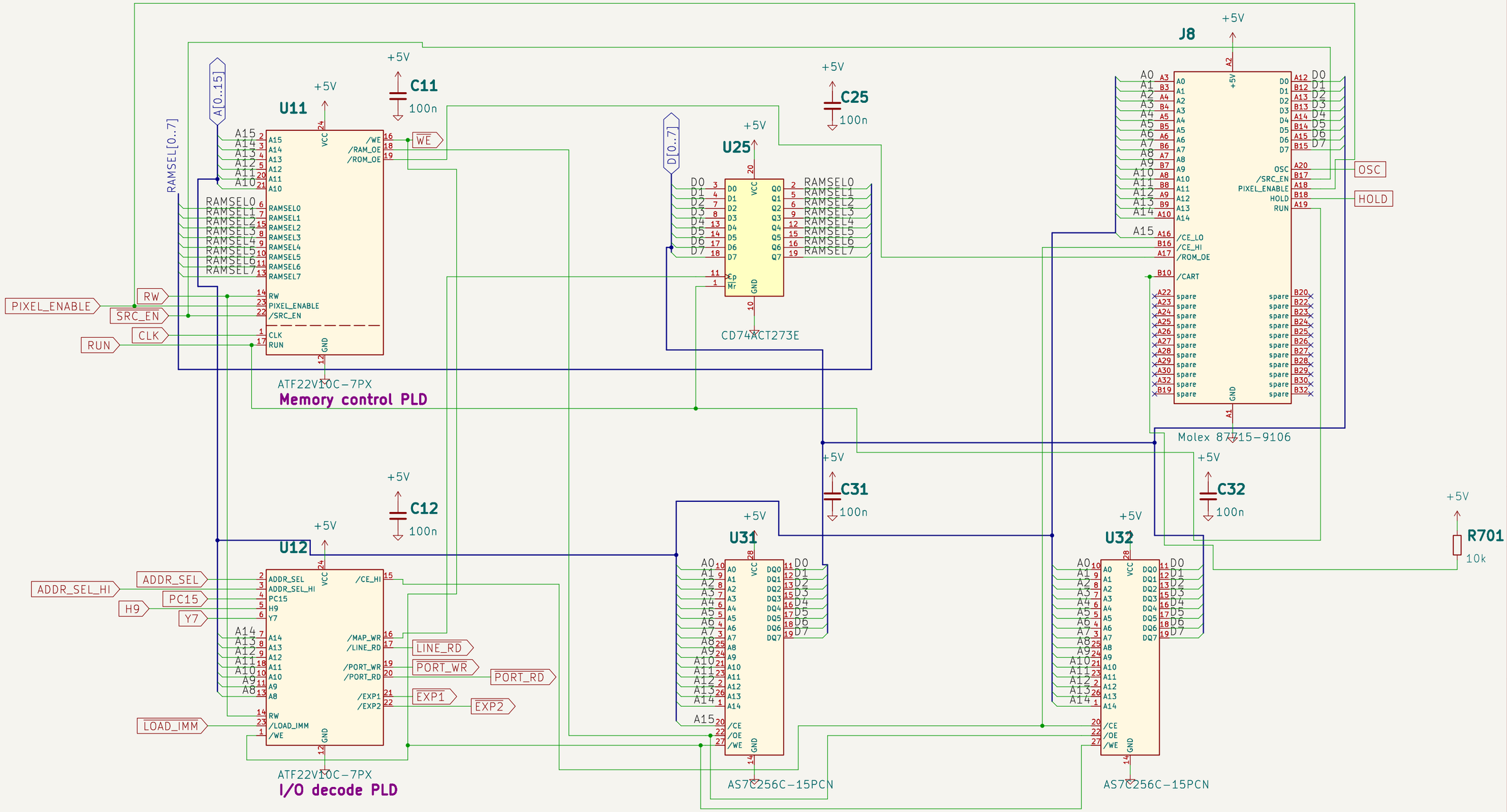
SRC7:SRC0: the two source F541s' joined outputs.
R601–R608 (47 ohm) limit the current while the
bus changes hands at the start of EXEC (CPU.md).
U26 drives D (sources 00, 01: /OE2 = I1),
U27 drives AC (10, 11: /OE2 = /AC_SEL).
RN601: D7:D0 read FFh when nothing drives them
(empty slots, the cartridge loader's feed byte).

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Data bus/	
File: data_bus.kicad_sch	
Title: Placard: Data bus	
Size: A3	Date: 2026-10-20
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: 1**
	Id: 6/11

Memory and I/O decode

RAM, ROM cartridge slot, memory control, I/O decode, map register



Lower 32K: /CE = A15 (RAM U31, cartridge /CE_LO); upper: /CE_HI from U12.
RUN low (cartridge loading): /ROM_OE held off, map register clear (all ROM).
/CART reads low with a cartridge in; nothing reads it yet. 22 spare contacts.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md

Placard computer

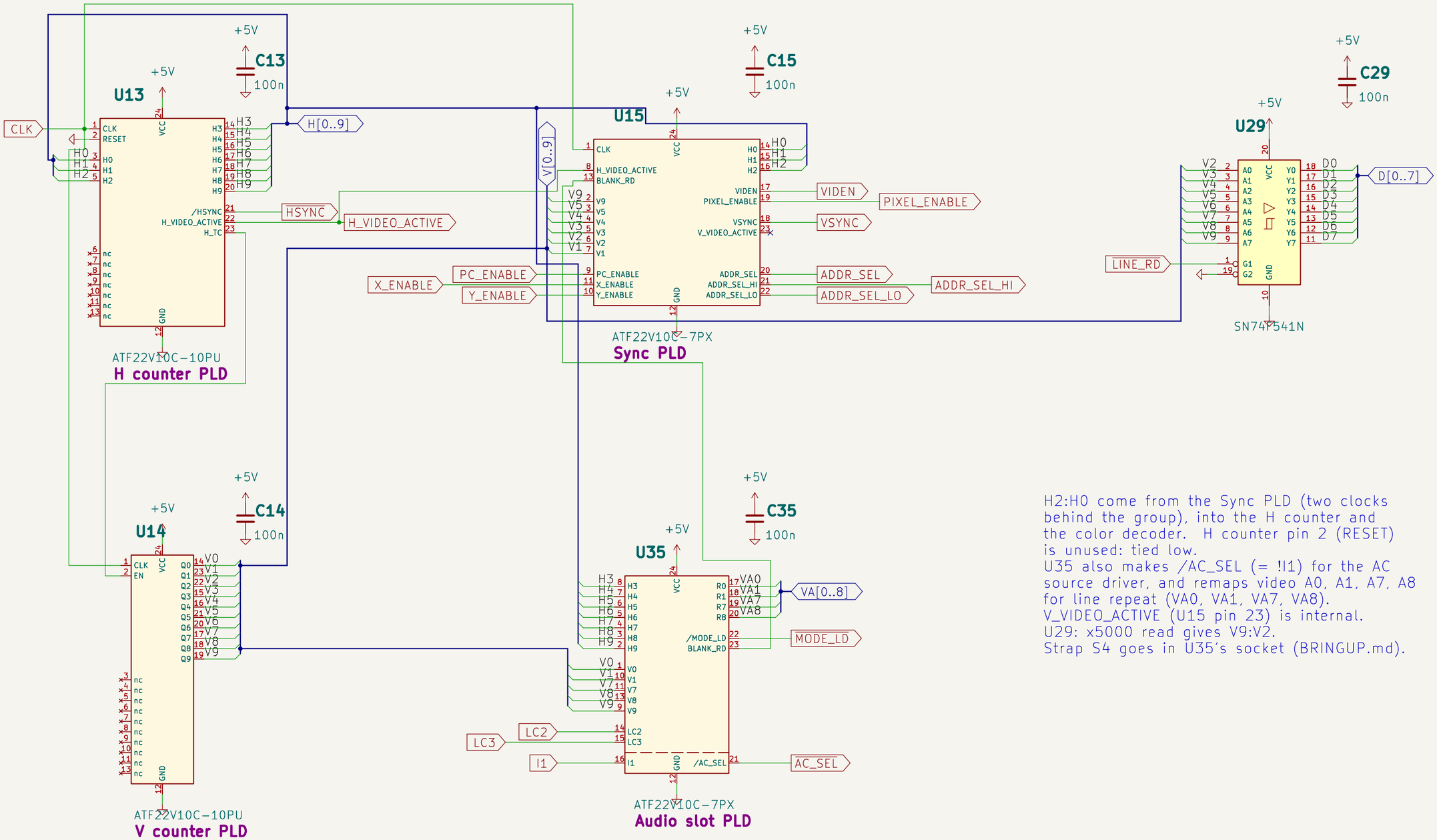
Sheet: /Memory and I/O decode/
File: memory.kicad_sch

Title: Placard: Memory and I/O decode

Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/11

Video timing

640 x 480 at 60 Hz: H and V counters, Sync, mode and audio slots, line read



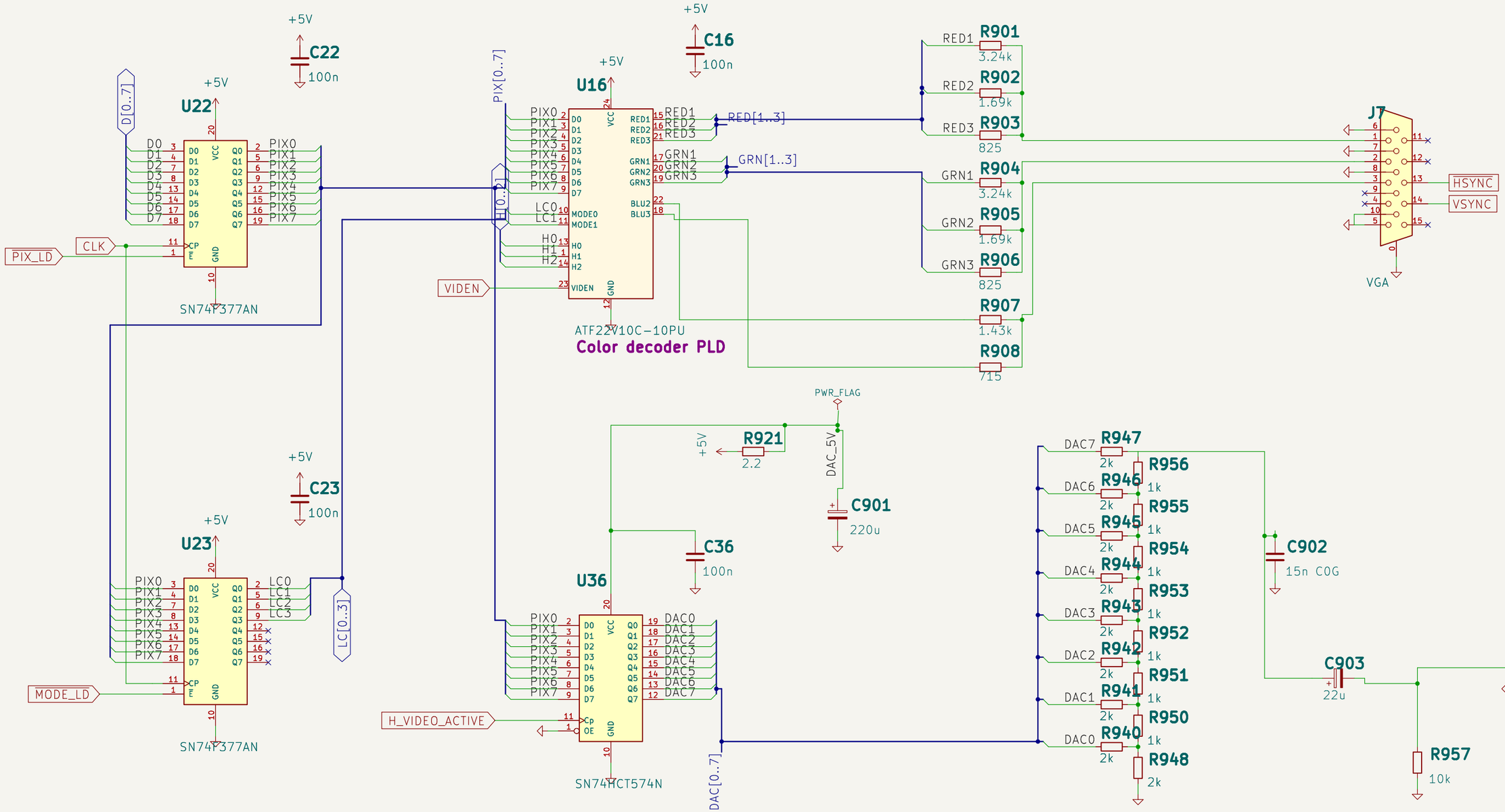
H2:H0 come from the Sync PLD (two clocks behind the group), into the H counter and the color decoder. H counter pin 2 (RESET) is unused: tied low.
U35 also makes /AC_SEL (= !I1) for the AC source driver, and remaps video A0, A1, A7, A8 for line repeat (VA0, VA1, VA7, VA8).
V_VIDEO_ACTIVE (U15 pin 23) is internal.
U29: x5000 read gives V9:V2.
Strap S4 goes in U35's socket (BRINGUP.md).

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Video timing/		
File: video_timing.kicad_sch		
Title: Placard: Video timing		
Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/11

Video and audio out

Pixel buffer, line code, color decoder and VGA DAC; audio DAC register, R–2R ladder, headphones



U23 Q3:Q2 = line repeat to U35 (sheet 8); Q1:Q0 = color mode. DAC_5V: U36's own filtered rail (2.2 ohm, 220u, C36).
Ladder: 2k 1 % per bit + 2k termination, 1k between nodes (LAD7 = out). AGND (GNDA) joins GND at the power input.

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

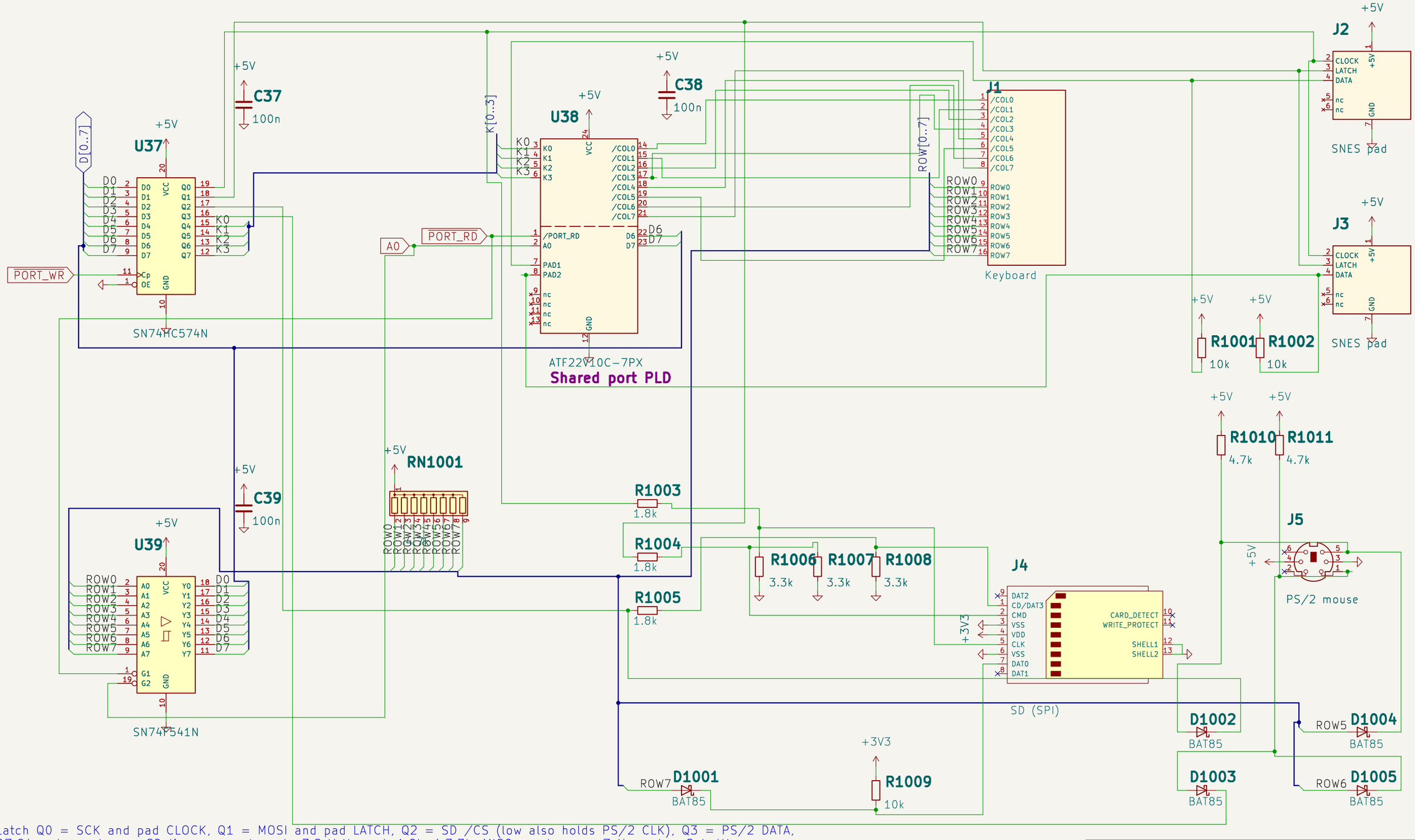
Sheet: /Video and audio out/
File: video_out.kicad_sch

Title: Placard: Video and audio out

Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/11

Shared port x5200

Keyboard, two SNES pads, SD card (SPI), PS/2 mouse



Latch Q0 = SCK and pad CLOCK, Q1 = MOSI and pad LATCH, Q2 = SD /CS (low also holds PS/2 CLK), Q3 = PS/2 DATA, Q7:Q4 = key column. SD lines come down to 3.2 V through 1.8k / 3.3k; MISO reaches row 7 through a Schottky.

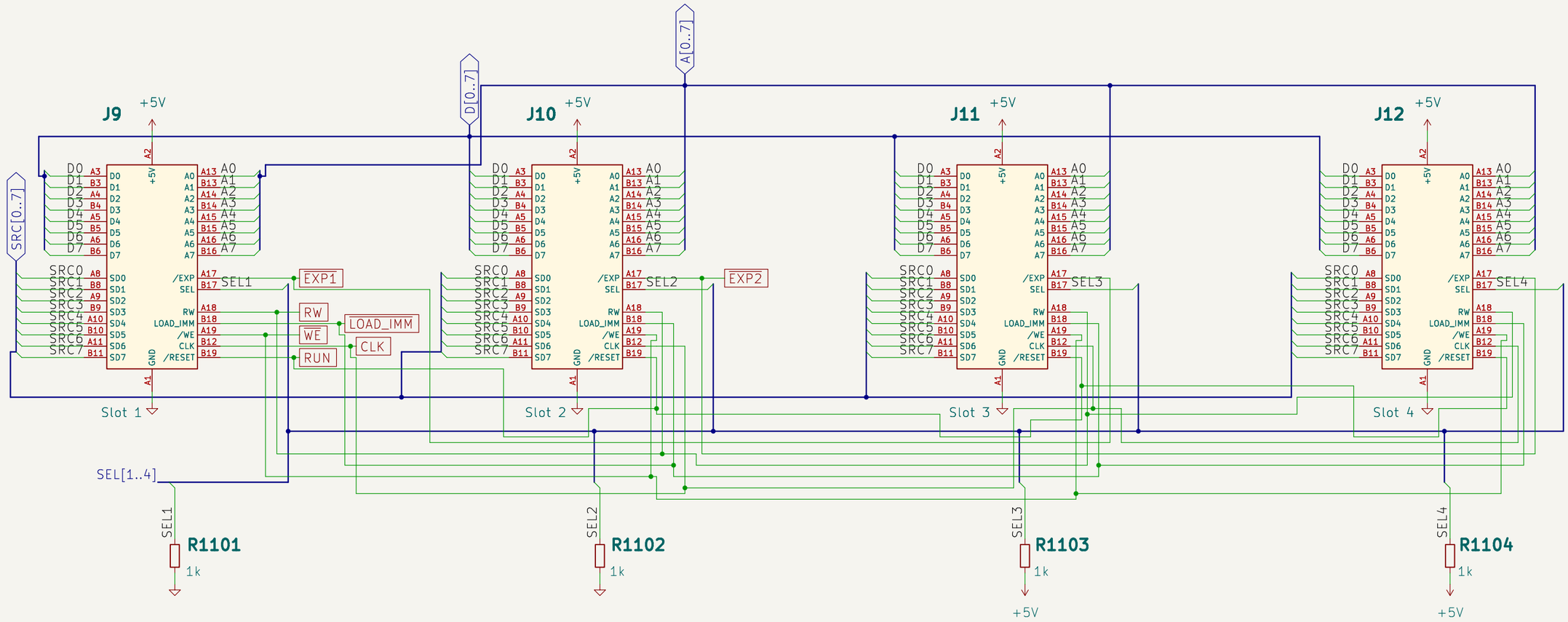
Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Shared port/	
File: shared_port.kicad_sch	
Title: Placard: Shared port	
Size: A3	Date: 2026-10-20
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 10/11

Rev: 1

Expansion slots

Four 2 x 20 card-edge slots, x5100–x53FF, 128 bytes each (EXPANSION.md)



SEL: GND on slots 1 and 2, +5 V on slots 3 and 4 (through 1k): the A7 half the card answers.
/EXP1 goes to slots 1 and 3, /EXP2 to slots 2 and 4. /RESET = RUN (low while the cartridge loads).
SD7:SD0 = SRC7:SRC0, the source drivers' side of the series resistors (the write data).
LOAD_IMM carries the State PLD's /LOAD_IMM (low when the CPU latches the read bus into D).

Chip numbers U1–U41 = docs/CHIPS.md; decoupling Cn goes with Un
Generated by src/scripts/gen_kicad_schematic.py; plan in docs/SCHEMATIC_PLAN.md
Placard computer

Sheet: /Expansion slots/
File: expansion.kicad_sch

Title: Placard: Expansion slots

Size: A3	Date: 2026-10-20	**Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/11